Which factors actually matters in student exam performance?

A theory-grounded, evaluation-driven visualization study of the StudentPerformanceFactors dataset (Kaggle)

Module: Fundamentals of Data Visualization (GDV)
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Semester: FS
Date: June 2026

Inhalt

Dataset Description

Where does the dataset come from?

The StudentPerformanceFactors dataset was taken from Kaggle (lainguyn123, n.d.). It is a synthetically generated dataset, built to simulate realistic student performance data across a range of demographic and behavioral factors.

Why this topic?

The dataset has many variables, but showing all of them would not give a clear answer to the research question. I picked three sub-topics, each covering a different type of factor and using a chart type that fits the data.

Sub-topic 1: Learning Habits I picked Attendance because it has the strongest correlation with Exam_Score (r = 0,58).

Sub-topic 2: Social and Family Factors I picked Family_Income to cover the social and economic side.

Sub-topic 3: Personal Motivation I picked Motivation_Level for the mindset side.

They cover the main types of factors: behaviour, background and mindset. Together they answer the research question by showing which type of factor matters most.

Which variables are relevant?

The dataset has 6,607 rows and 20 columns. The target is Exam_Score (numeric, 55 to 101). The input variables we focused on:

Numeric:

- Hours_Studied (1 to 44 hrs)
- Attendance (60 to 100%)
- Previous_Scores (50 to 100)
- Tutoring_Sessions (0 to 8)
- Sleep_Hours (4 to 10)

Categorical (all Low / Medium / High):

- Parental_Involvement
- Motivation_Level
- Family_Income
- Teacher_Quality (78 missing values)
- Access_to_Resources

Data Preprocessing

Starting point

Raw dataset: StudentPerformanceFactors.csv from Kaggle (synthetic). Initial shape: 6,607 rows × 20 columns. Three columns contained missing or empty values: Parental_Education_Level (90), Teacher_Quality (78), and Distance_from_Home (67). Every row containing at least one missing or empty value was removed. This dropped 229 rows in total, leaving 6,378 rows × 20 columns after cleaning. Listwise deletion was chosen over imputation because the dataset is synthetic and large enough to absorb the loss: the 229 removed rows account for only 3.5% of the data, which keeps any introduced bias minimal. A note on the score range The maximum Exam_Score is 101, which is slightly unusual for a typical 0–100 scale. Since the dataset is explicitly synthetic, this was accepted as-is and no further filtering was applied. The single high value has a negligible effect on the distributions and correlations.

Transformations

For the correlation analysis only (not for the visualizations), the categorical variables were encoded as ordinal integers. This encoding was used solely to compute Pearson correlations with Exam_Score. The original string values were kept in the cleaned dataset and used for every visualization. The cleaned data was saved as data/data_cleaned.csv and served as the input for all subsequent visualizations.

Design Rationale & Theory

Chart 1: Exam Score by Motivation Level (Sub-topic 3: Personal Motivation)

What it shows

The chart displays the full distribution of exam scores for three motivation groups: Low, Medium, and High. Each box shows the interquartile range (25th–75th percentile), the line inside each box shows the median, the whiskers extend to 1.5 × IQR, and individual outliers appear as separate dots. The goal is to show not just whether average scores differ between motivation groups, but whether the distributions overlap substantially, something a bar chart would hide. The independent variable (Motivation Level) is ordinal categorical and the dependent variable (Exam Score) is numeric. A bar chart was deliberately avoided here. The correlation between Motivation Level and Exam Score is weak ($r \approx 0.15$), which means the group differences are small relative to the within-group variance. Showing only the mean would give a misleadingly clean picture. The boxplot makes the overlap between groups visible and is therefore more honest to the data. The resulting chart is shown in Figures: Chart 1 (before evaluation).

Theory

The boxplot was introduced by Tukey (1977) as a tool for communicating distributional shape efficiently. It encodes five summary statistics at once: minimum, lower quartile, median, upper quartile, and maximum (within the whisker range). The median line uses position on a common scale, the most accurate encoding channel in Cleveland & McGill's hierarchy (1984), which lets readers compare medians across groups accurately. The Gestalt principle of similarity (Munzner, 2014) explains why the three boxes are perceived as a comparable group: identical width, aligned on a common baseline, placed on a shared axis. This grouping supports direct comparison without additional visual cues.

Chart 2: Average Exam Score by Family Income (Sub-topic 2: Social & Family Factors)

What it shows

The chart compares the mean exam score across three family income levels: Low, Medium, and High. The x-axis shows the income category in ascending order, the y-axis shows the average exam score, and numerical labels sit on top of each bar so values can be read precisely. The goal is to show whether socioeconomic background, measured through family income, is associated with higher or lower exam performance. The independent variable (Family Income) is ordinal categorical with three levels. The task is to compare a single summary statistic across groups, which is the primary use case for bar charts. The three categories follow a natural order (Low - Medium - High) that aligns with the left-to-right reading direction, making the trend readable without additional explanation.

A boxplot would be an alternative, but since the research question focuses on group-level differences rather than distributional shape, the bar chart communicates the comparison more directly. A line chart would be inappropriate, because Family Income is not continuous and the points between categories carry no meaning. The resulting chart is shown in Figures: Chart 2 (before evaluation).

Theory

Munzner (2014) defines the appropriate chart type based on the combination of data type and analytical task. For the task compare with categorical and quantitative data, bar charts and dot plots are the canonical choices. Bar charts rely on the length encoding channel, which Cleveland & McGill (1984) rank second in perceptual accuracy after position on a common scale. Tufte's principle of maximizing the data-ink ratio (Tufte, 1983) supports the minimal styling applied here: no gridlines on the x-axis, no chart border, and value labels that remove the need for detailed y-axis reading.

Chart 3: Attendance vs. Exam Score (Sub-topic 1: Learning Habits)

What it shows

The chart displays the relationship between a student's attendance rate (x-axis, in %) and their exam score (y-axis). Each dot represents one student from the cleaned dataset (n = 6,378). The goal is to make the correlation between attendance and performance visible and to check whether the trend holds consistently across the full range of attendance values. Both variables are numeric and continuous. A scatterplot is the natural choice for showing the relationship between two quantitative variables, since both values can be mapped to position on a common scale, one for each axis. According to Cleveland & McGill (1984), position on a common scale is the most accurately perceived visual encoding available, which makes the scatterplot the strongest fit for correlation tasks. Alternatives such as a heatmap or binned density plot would aggregate the data and hide individual variation, which is not appropriate here because the spread of the data is itself informative. The resulting chart is shown in Figures: Chart 3 (before evaluation).

Theory

Cleveland & McGill (1984) established a hierarchy of perceptual accuracy for visual encodings. Position on a common scale sits at the top, above length, angle, area, and color intensity. By encoding both Attendance and Exam Score as positions, the scatterplot maximizes the reader's ability to perceive the relationship accurately. The Grammar of Graphics (Wilkinson, 1999) formalizes the mapping: x = Attendance, y = Exam_Score, mark = point. No additional aesthetic channels are needed, because the message is a bivariate relationship between exactly two variables.

Evaluation

Method

A small Think-Aloud study was used to evaluate the three charts. Each participant was shown the visualizations one by one and asked to narrate what they noticed and what they understood, while the moderator took notes. Think-Aloud was chosen because it is a lightweight, formative method that surfaces interpretation problems early without requiring a large sample, which fits the scope of this project.

Participants

Three participants took part in the evaluation: two fellow students and one family member and a friend of mine. They represent a small range of visualization literacy, from "familiar with charts in an academic context" to "non-technical".

Results

Chart 1: Motivation Level

Most participants noticed straight away that the three motivation groups look very similar. Some pointed out that the high-motivation group seems to score slightly higher than the others, but they did not perceive the difference as strong. Several mentioned that they had to look at the chart for a moment before the main message clicked. The boxplot was useful because it showed the distribution of exam scores, but not everyone found it easy to read at first. The outliers above and below the boxes were mentioned as slightly confusing, since it was not immediately clear that those dots represent outliers.

Overall, participants understood that motivation has some connection to exam performance, but they also felt that the chart needed a clearer explanation. Suggested improvements included a short note below the chart explaining how to read the boxplot, and a more descriptive title that states the actual finding (for example, that higher motivation is linked to slightly higher exam scores).

Chart 2: Family Income

Most participants found this chart easy to understand, since the bar chart is simple and the values are shown directly above the bars. Several people quickly noticed that the average exam score increases from low to high income. However, some participants also said that the difference between the bars looks bigger than it actually is, because the y-axis starts at 60 instead of 0. This made the visual impression slightly misleading. A few participants also pointed out that the chart only shows the average score, without indicating how much the scores vary within each income group.

Overall, the chart was seen as clear and readable, but the message needed to be presented more carefully. Suggested improvements were to indicate that the difference is small, and to use a boxplot or add some information about the distribution so the reader sees more than just the average.

Chart 3: Attendance vs Exam Score

Several participants said this was one of the clearest charts. The first thing they noticed was that the points generally move upward, which suggests that higher attendance is connected to higher exam scores. Some found this chart particularly useful because the pattern was easier to see than in the other two. However, others mentioned that the chart looks quite full, with many points sitting close together. In some areas the points overlap, which makes it hard to tell how many students are actually represented. A few participants also said that a trend line would make the message even clearer.

Overall, the attendance chart was seen as strong and understandable, but it could look cleaner. Suggested improvements were to make the points smaller and more transparent, and to add a trend line so the positive relationship is visible at a glance.

Overall evaluation

Across the three charts, the feedback was that the visualizations were understandable, but several parts could be polished to make the message clearer. The attendance chart was often singled out as the strongest, because the relationship was visible almost immediately. The family income chart was seen as the easiest to read, since bar charts are inherently simple, but it risked exaggerating the differences visually. The motivation chart was useful but the most difficult for some participants, since boxplots are not immediately clear to everyone. The main improvements identified across the three charts were clearer, finding-driven titles, short reading guides below the charts, a trend line for the attendance scatterplot, more transparent points to reduce overplotting, and a more honest presentation of small differences.

Design Improvements

What was changed after the evaluation, and why?

Chart 1: Motivation Level (Boxplot)

Feedback: Participants understood the boxplot, but some found the outliers confusing and the main message unclear.

Changes:

- The title was rewritten to communicate the actual finding: "Motivation Level Shows Only a Small Effect on Exam Score."
- A short caption was added below the chart explaining how to read the boxplot (median, IQR, whiskers, outliers).
- The color palette was changed from red/orange/green to a neutral single-hue scheme, to avoid implying a value judgment.

Rationale: A descriptive title removes ambiguity about the message, in line with Munzner's (2014) principle that visualizations should communicate the analytical insight directly. The caption supports readers who are unfamiliar with the boxplot convention. Neutral colors avoid encoding extra meaning that the data does not support. The revised version is shown in Figures: Chart 1 (after evaluation).

Chart 2: Family Income (Bar Chart)

Feedback: The chart was easy to read, but the truncated y-axis exaggerated the differences, and the chart hid the variance within each group.

Changes:

- The y-axis was extended to start at 0, giving an accurate visual impression of the differences.
- A short note was added stating that the differences between income groups are small (range: 66.9 to 67.8).
- The red/orange/green colors were replaced with a single neutral color.

Rationale: Tufte (1983) warns against axis manipulation that distorts proportions. Starting the y-axis at 0 restores the visual honesty of the chart. The added note prevents the reader from overestimating the effect. A single color is sufficient here, because the categorical labels already encode the groups, so additional color becomes chartjunk. The revised version is shown in Figures: Chart 2 (after evaluation).

Chart 3: Attendance vs Exam Score (Scatterplot)

Feedback: The strongest of the three charts, but overplotting reduced clarity and the trend was harder to see than expected.

Changes:

- Point size was reduced and transparency was increased (alpha = 0.3) to address overplotting.
- A linear regression line was added with a 95% confidence band.

Rationale: Transparency lets the reader perceive point density, which carries information about how many students share similar values. The regression line makes the direction and strength of the relationship immediately visible, supporting Cleveland & McGill's (1984) principle of accurate perception. The revised version is shown in Figures: Chart 3 (after evaluation).

Figures

Chart 1: Motivation Level

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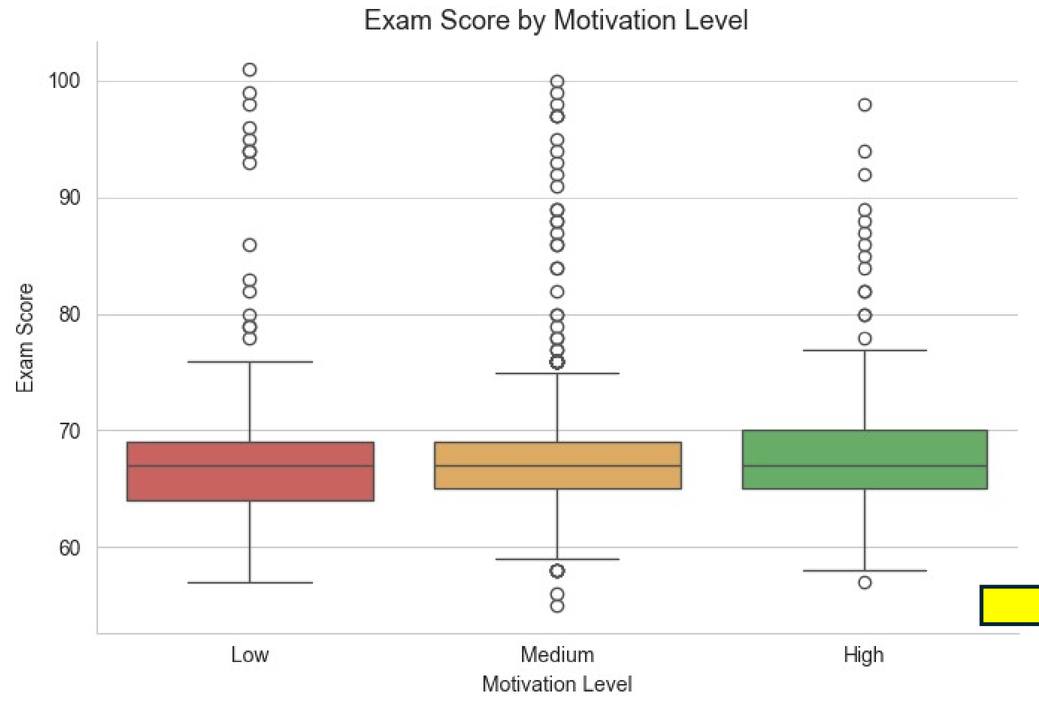
<tr><td>


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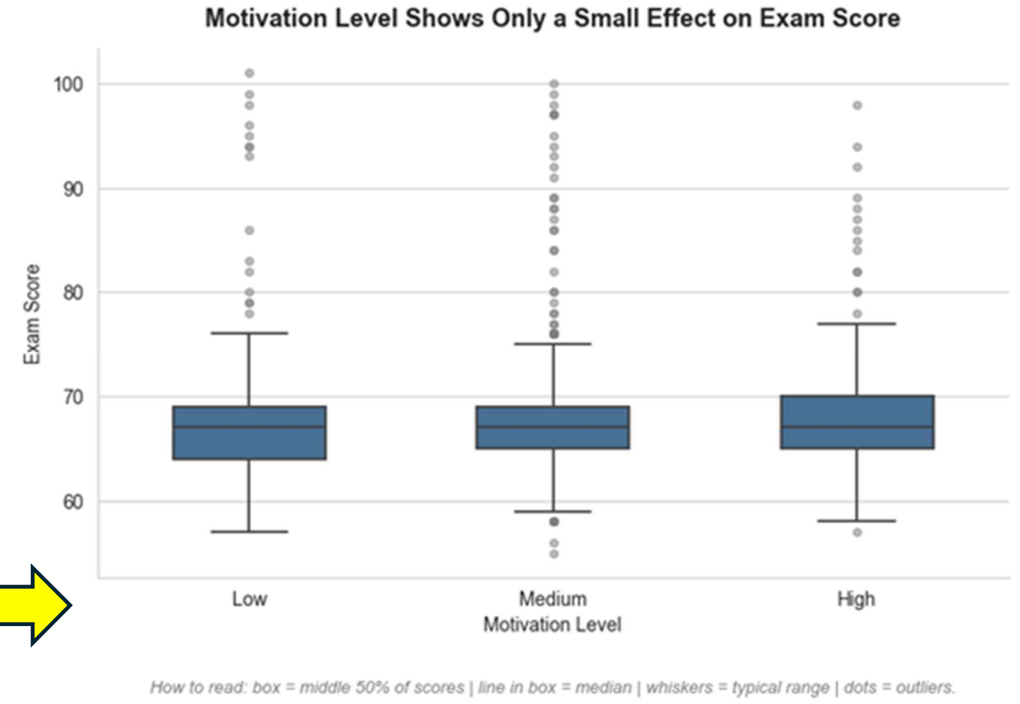
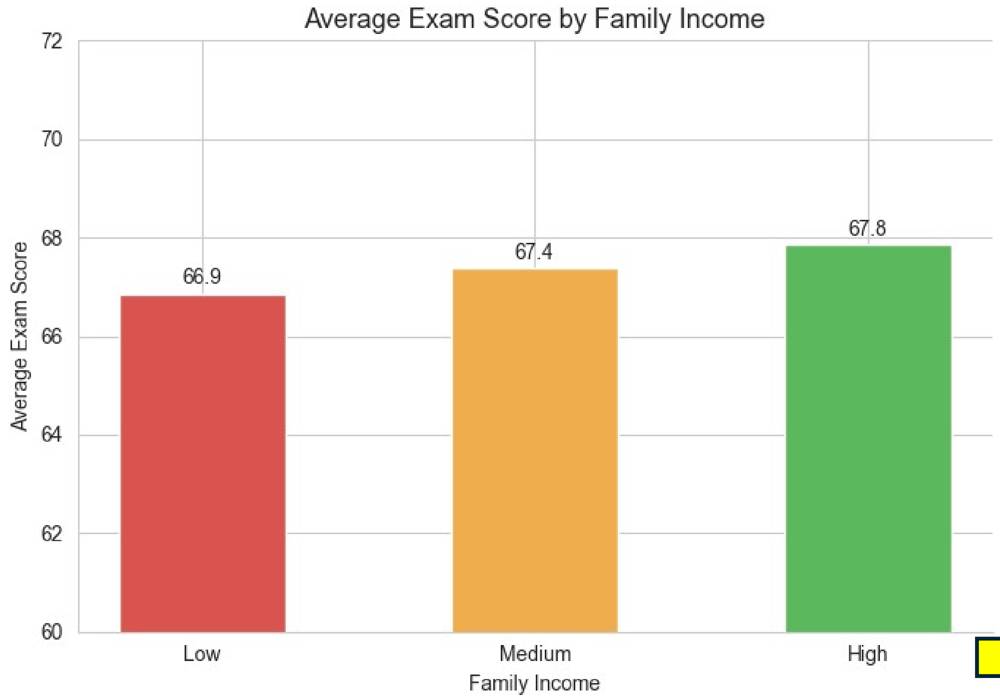


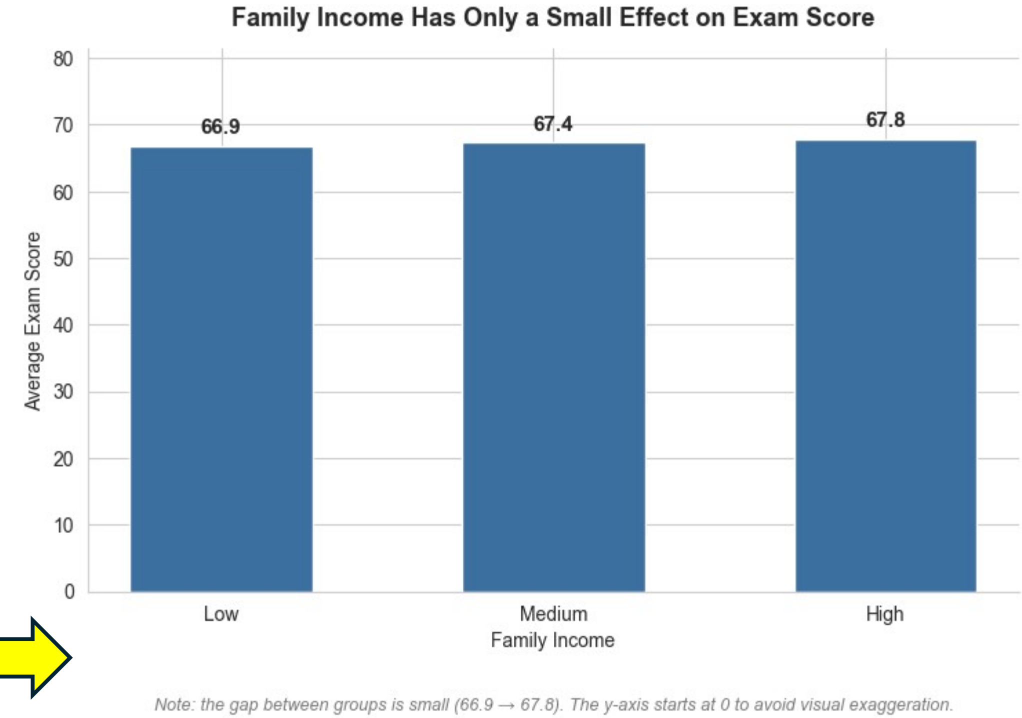
</td></tr>
<tr><td colspan="2">What Changed:
- Red/orange/green palette replaced with single neutral blue (avoids value judgment)
- Added reading guide caption below chart explaining how to interpret boxplot elements
- Rewritten title to communicate the actual finding: "only a small effect" not just</td></tr>
</table>

Chart 2: Family Income (Bar Chart)

Vor Evaluation



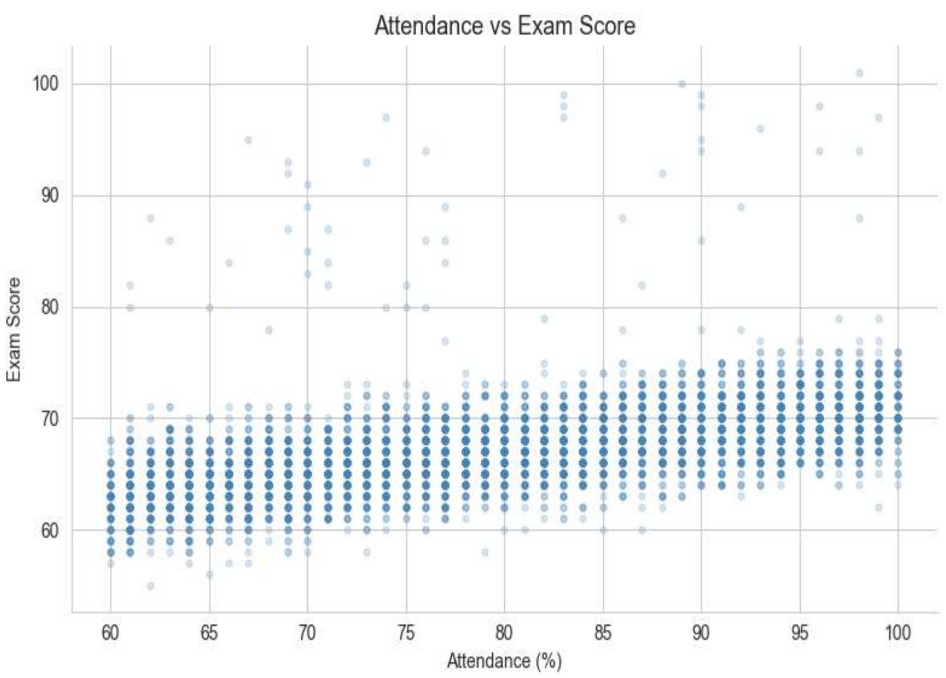
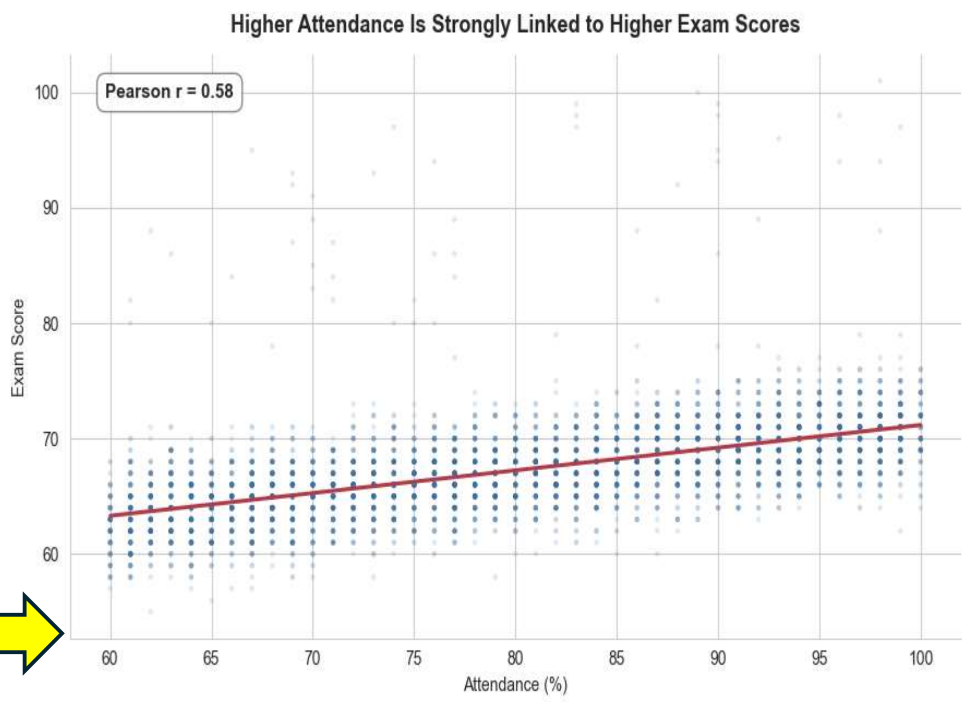
Nach Evaluation



What Changed:

- Y-axis now starts at 0 (was 60) to avoid visual exaggeration
- Single neutral blue replaced red/orange/green to remove color judgment
- Added caption explaining the small difference (66.9 vs 67.8)

Chart 3: Attendance vs Exam Score (Scatterplot)

Vor Evaluation	Nach Evaluation
	

What Changed:
- Red/orange/green palette replaced with single neutral blue (avoids value judgment)
- Added reading guide caption below chart explaining how to interpret boxplot elements
- Rewritten title to communicate the actual finding: "only a small effect" not just variables

Insight

Chart 1: Motivation Level vs Exam Score

Motivation level shows almost no effect. The three boxes overlap almost completely and the medians are nearly identical. The High group has a slightly higher median, but the spread within each group is much larger than the difference between groups. A bar chart would have hidden this, but the boxplot shows that motivation as measured here does not separate students in any meaningful way.

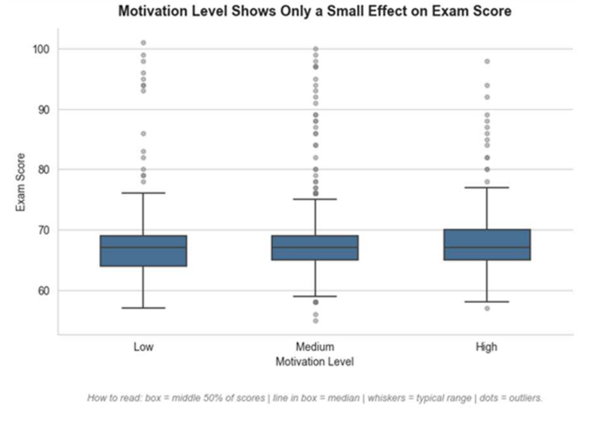


Chart 2: Family Income vs Exam Score

Family income has only a small effect on exam performance. The three groups score 66,9, 67,4 and 67,8 on average, which is a difference of less than one point. The direction matches the expectation that higher income helps a bit, but the size of the effect is too small to call income a main driver of performance. The honest y-axis starting at zero makes this clear at one glance.

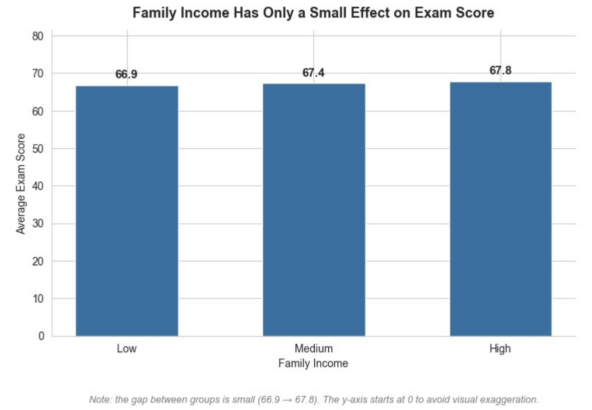
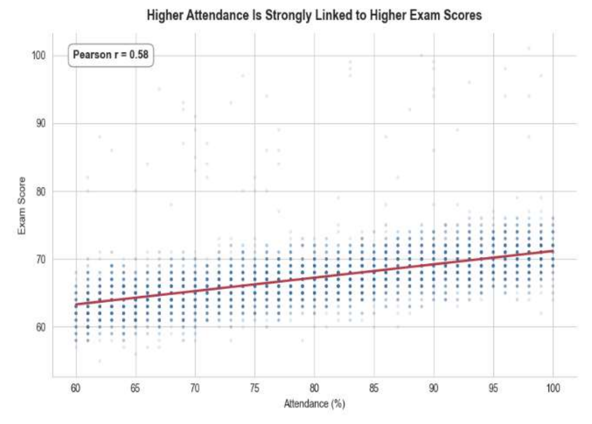


Chart 3: Attendance vs Exam Score

Attendance is by far the strongest factor in the dataset. The regression line shows a clear positive trend, and the narrow confidence band means the relationship is stable across all students. Students who attend more classes score consistently higher, with a correlation of r = 0,58. This is the only factor in the three sub-topics with a real, visible effect.



Across the three sub-topics, behaviour beats background and mindset. Attendance is the dominant factor, while family income and motivation level only show small differences. The answer to the research question is therefore clear: what students actually do matters far more than where they come from or how motivated they say they are.

References

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